

ActInf Livestream #034.1 ~ "The free energy principle: it's not about what i

Livestream | 1:43:49

[Music]

hello and welcome everyone to actin flab

live stream number 34.1

it's december 7th 2021.

welcome to actinflab

we are a participatory online lab that
is communicating learning and practicing
applied active inference you can find us
at some of the links

here on this page

this is a recorded and an archived live
stream so please provide us with
feedback so that we can improve on our
work

all backgrounds and perspectives are
welcome here and we'll be following good
video etiquette for live streams

all acting flap activities are
participatory so let us know if you want
to join in or co-organize we have
three organizational units education
communication and tools and weekly
meetings for each of those units a ton
of interesting projects to get involved
in

coda.io at activeinferencelab activelab
you'll find a searchable and sortable
site where you can learn more about the
past and upcoming live streams that we
have

today in acting stream number 34.1 we're
going to be learning and discussing
about this

fun paper the free energy principle it's
not about what it takes it's about what

took you there by actual constant in

2021

and in

34.0

dean and i had a great time we talked

about the paper so check out the dot

zero

and of course read the paper if you want

to learn more

today in 34.1 we can go over some of the

key claims

we can look at some of the examples

again and differentiate the bayesian

approach versus the free energy approach

and just really take it anywhere that

people think is interesting so if you're

watching live

just

ask questions in the live chat and we'll

be keeping our eye on that and we will

just introduce ourselves go around and

then we'll

take it wherever we take it

so

we can introduce ourselves we can say hi

and say maybe something that was

exciting about the paper

that we remembered or something that

we'd like to reduce our uncertainty on

by the end of this discussion

so i'm daniel i'm a researcher in

california

and

i'm actually just looking forward to

seeing if anyone in the live chat has

some questions to ask because there's so

many cool adjacencies to this paper and

just the

beginnings of the thread were explored
by dean and i in the dot zero so
exploring the implications and what
people see as relevant in this paper i
think will be really exciting
and i will pass it to dean
thanks daniel i'm dean i'm calgary um
i'm just curious again to see how
as you start pulling on a thread
discovering how many things are actually
there as opposed to just the threads so
yeah i think today should be a pretty
interesting day and i'll pass it over to
david

i'm dave i'm

live north of manila

i did natural language processing with
big computers and

studied cybernetic learning theory and
i'm

excited that he's given us some simple
math actual axle constant is kind of
bringing it down to those of us who
stalled by the end of calculus one
just to sort of

lead-in question dave in the area of
cybernetics

is there any

analogous work anyone who said well it's
not really about being a cybernetic
system it's about how you came to be
that cybernetic system like are there
any parallels

in the cybernetics literature for the
kinds of things that are explored in
this paper

well there are yeah um in terms of um

[Music]

not so much necessarily the origins well
yes in cybernetic learning systems this
theory um a branch called
conversation theory
uh they completely cut loose the
notion of the physical embodiments of
systems from the
mimetic
or functional or conceptual or cognitive
aspects of systems in fact even types of
systems uh very much like what richard
dawkins did
30 plus years ago on separating
genetics from my medics which in his
case originally was just a proof of
principle not something he wanted to
pursue especially and it has been
pursued um in cybernetic learning theory
um the individual
is equated to the conversation
is equated with a s a type of system
that originates that persists
that merges into other systems
that may very well function through many
bodies
so you have distributed intelligence
right out of the gate
um
as well as internal differentiation
internal self-definition of each
individual as a conversation
so
definitely
the
there and there's a lot of math even
going way back to the 60s and 70s they
they
uh

there were cyberneticians who took
precise quantitative work very seriously
and dug down
uh very carefully on quantified
attention quantified uncertainty
resolution of uncertainty and the
qualitative
uh
consequences
of deciding
what's really happening what my plan
really is very much uh the sort of
fun
abrupt and
often irreversible changes that
uh
the world looked at in a lot of depth
when catastrophe theory burst upon the
world sudden catastrophic perhaps an
improvement but a catastrophic a
suddenly relatively irreversible change
um
[Music]
and here it is still working on it in
deeper and deeper in detail and wider
scope so good for mr dr cost
pretty
interesting it makes me think about like
a friendship
and how
maybe it's not just about
what the friendship is
or what it takes to have a friendship
but it's about the specific
conversations that take you there
so let's just kind of
look at the main
aims and claims and parts of the paper

and then either of you raise your hand
of course
maybe some others will join and anyone
in the live chat please just ask any
question or
make a comment we're just chilling
so the center
topic that's raised and it's a term
that's introduced in this paper
uh not the word but its usage in this
situation is the entailment problem
and just like the dog and the tail as
they say
the entailment problem is about
the relationship the
necessity and the sufficiency
relationship which ones following which
and under what situations they're
following each other
the entitlement problem is the confusion
in the entailment relation between free
energy minimization and life so
is free energy minimization
necessary for life is it sufficient
those are
the core pieces of the entailment
problem
and what happens in this paper
is a
simple dissolution through a trivial
toy
simplified example
of that entailment problem and we're
gonna probably walk through it again
today
but this numerical example which is very
stripped down it's not full active
there's no action in the loop it's just

a perceptive example
and it just shows that free energy
minimization does not always lead to the
correct choices
it's also conditional upon the priors
in other words how you got there it's
about what took you there because the
priors don't come from nowhere they come
from the past
so this kind of connects this
uh rigorous logic
within the action perception loop or in
this case just a simplified perception
loop to the
often ill or a logical way that things
happened to get the way they are so it's
going to be that
tension between
is this a rational analysis is it
locally rational maybe globally
irrational all those kinds of topics
come up and just to
skip to the final words of the paper
the claim
in this paper that's explored is that
free energy minimization is not
sufficient for life so not that fep has
nothing to say about life not that
actinf is irrelevant but that free
energy minimization is not sufficient it
turns out that it's necessary to be
doing something like free energy
minimization to persist in a dissipative
world but it's not sufficient it can't
be all that you have the wheels are
necessary for the car to go round and
round round and round but they're not
sufficient

and then that last sentence
the free energy principle is meant to
account for all kinds of systems
you the listener slash reader
the author of the paper
and the system of interest the organism
under study in a unifying fashion so we
should also explore like what is the
unifying
lens or what is the framing
under which
the research scholar and the person
who's just sort of like what is the free
energy principle and the bacterium and
what is that unifying principle and what
does the free energy principle have to
do with that
okay
so
dean where do you think we should jump
in or what would be um a fun question or
place to start here
well so i have
i have an interest
a little bit history with this so what
what took you there so if i'm meeting
somebody and they're an expert at
something i don't typically ask why are
you an expert because that's kind of
offensive but to ask what
what got you here what got you to the
place where you now
bear this title
the weight of that title isn't is an
interesting way of going about
trying to figure out
what
what logics they have applied in the

past
to
untangle problems and resolve things
that may be somebody
with less expertise
struggles with her or doesn't seem to be
able to overcome
so that's where this paper was really
helpful to me
instead of
why do i have to do this or why are you
such an expert
asking where did you come from
and then when being specific in terms of
so
how did you resolve something that i
still find
difficult to resolve like why do i have
to tie my shoes and then the expert says
well because you're going to trip over
your shoelace and fall down well i
haven't tripped over my shoelace instead
of going down that path
figuring it out in a slightly more
sophisticated way
cool thank you um i'm not gonna read
this blate quote but of course we have
to write it down
it makes me think about how why
which is an important question
often lends itself towards
generalization like why are you studying
machine learning well it's really
important and it's going to become
increasingly important and i wanted to
have a good job and other people told me
it was relevant it kind of generalizes
out

whereas what took you there is always going to be a trajectory it's kind of a path and that's where you get the answer that's more like

well i met this mentor and then i was curious about this specific topic and then we did this program

so

what took you there is about the specifics and

what

is broached in why questions is often generalizations and we know that there's not even a single answer to why like we've talked about aristotle's whys and about tin bergen why like why is always going to be a plurality of answers

whereas what took you there is going to be a specific path and so that's the tension

in living systems

is we want to understand something general like surely there are some generalizations we can make across single-celled life or across mammals but

also we're explaining specific systems

so

which

aspects are going to be generalizable and which ones aren't and i think again free energy principle as a theory of everything

and something that does clearly make generalizations across systems at least apply to different systems

how does this

hyper

specific

trajectory based way of approaching

answering a question

relate to more of a mean field

approximation

averaged over

multiple categories

i think it's that tension that's really

interesting to describe so one is this

general versus specific tension

and then

if we can go to the simple point that's

made in the paper

that the free energy principle is not

concerned with the sufficient conditions

of existence but rather with what must

have been the case given that you exist

so this is related to necessity and

sufficiency which we talked about in the

dot zero they're similar words so we

should use them differently they're

different words

in other words it's about necessity but

not

necessarily sufficiency

and then it's not about what figuring

out what it takes to be alive it's about

figuring out what took you there

so

the title

is restated and turned around many many

different times because it is such a

distillable and simple yet important

concept and i think the piece that

i'd like to explore a little more is

really the pronoun use

i understand why the paper is written

from an i perspective because that's
convention with single author philosophy
papers
but
it's about figuring out what took you
there
so there's a sort of rhetorical element
of speaking to a reader which is
returning at the end of the paper those
last few lines about you me and the
system under study
but then also it's almost like
about how the free energy principle
is about systems looking at their own
timeline and trajectory
and perhaps not as much as looking out
so i'm curious about what does that
relational
element
mean here with the person relating to
their past and
us as systems relating to other systems
dean
yeah so the necessity might be you show
up
you attend in order for
things to
sort of continue on but the sufficiency
part is how available you are so you can
show up and turn all your senses off you
can be present but not necessarily
be taking in anything
or at least trying to actively avoid
taking information in and so
that's the part again i would agree with
you that's the part i found really
interesting because
to to not understand or appreciate that

there's a difference between what is
necessary

i'm i'm here i'm in part of this live
stream now but then i could be
distracted in all different directions
not really be able to be available
to what

is going on around me is is a whole
different matter and i think that's what
axel has done a really good job of
bringing it down to that personal level
sufficiency

has to be exhaustive
to say what's truly sufficient for the
bacteria to live

it it's going to include things like
well you need the laws of physics and
you need gravity and you need um just
everything it's like really you're
giving

almost a bit by bit play-by-play
so sufficiency

it requires an exhaustive account
because if you're missing something
that's necessary

you didn't capture what was sufficient
necessity allows us to just focus on
some aspect that we know is relevant
and then talk about

the why and the what and the how of
necessity

while acknowledging that we're just
focusing on one piece like that
one line in that one kandinsky painting
it's necessary for my interpretation of
it

it's not sufficient because the
sufficiency of the interpretation of art

would just send you off into the whole
universe maybe you need all of human
cultural history including from cultures
that you wouldn't even directly
acknowledge
for the sufficiency
to be
garnered in that interpretive setting
but necessity is a lot easier to point
your finger to
that relates to what
um
are distinguished as the strong and the
weak
responses to the entailment problem
so let's look over at this and again
anyone with any kind of a question
please just ask it because we're
chilling and
looking forward to your questions
so this is where in the paper the
entailment problem is
specified arguments in the literature on
the free energy principle give the
impression that in order to be alive
to count as a living system one must
minimize free energy
and so
there's a few ways to think about this
relationship between minimizing free
energy and life
sorry if you thought it was going to be
a discussion on a different topic
because that's what this paper is about
and it just again it's so simple it's
like
a shape in your hand you're turning it
over the shape is simple in some ways

but then the implications of the shape
and the way it gets
connected to other ideas is really
critical
and
actual distinguishes two
ways to talk about the relationship the
strong and the weak
and as we'll see
they differ not just in their strength
but also
essentially in the direction which the
arrow is pointing
so it's kind of like it's not like
there's a little mountain in a big
mountain but the arrow of elevation is
pointing the same way there's something
a little bit different happening
the strong
claim
in response to the entailment problem
is that minimizing free energy is a
sufficient condition for life and that's
also called in some earlier work the
overly generous claim which is kind of a
framing that already says that it's
probably
uh
too generous it's incorrect in that
way
and the strong claim is the sufficiency
claim
minimizing free energy that's all you
need it's sufficient to minimize free
energy to be a living system
the implication of the strong claim
would be that all we need to do is have
that free energy thermometer and then

we can determine who's minimizing free energy we have our meter stick on this left side and then

that would be sufficient to talk about

what systems were alive or not

in contrast the weak claim

is a necessity claim

if a system is currently alive it means

that it has minimized its free energy

and such a type of claim does not assume

the fep is designed to set the bar for

the sufficient conditions of life or

meant to predict what things may or may

not be alive

rather it limits the scope of the

application of the principle to beings

that we think are alive now

and enables us to know the necessary

conditions under which those beings can

be living i.e can actively resist the

loss

of structural integrity what took them

there

so

there's a few interesting pieces just

they differ in strength these two

responses or claims that are

generated by the entailment problem

i wonder what is the scope of the fep

and is it for systems that we have a

priori

decided are already alive

are

non-living systems under the scope of

the fep

so maybe we can try to like do some

duplicate the slide and do a little

diagramming and venn diagrams on

what
does the minimizing free energy
circle look like with respect to the
life circle and then what is life if we
don't have the thermometer over here on
the left side
so can i ask a question daniel
would it
would it be going too far off the rails
if i
if i wondered
to take this idea
and say okay so i want to build up i'm
i'm
i'm an academic or i'm a trainer and i
want to
i want to build out a curriculum
knowing this
what does that imply in terms of
well i know what's the difference
between something that's
surprising removal because i remember
i'm the one that already knows this and
i'm trying to pass this along to
somebody who doesn't know it so it's
i'm i'm trying to guess what surprisal
removal
or minimization means to them i'm
supposed to be guessing what life means
to them i'm supposed to be knowing what
is necessary what absolutely they must
know and what will be sufficient so in
that context
how would we um
[Music]
how would we order this
let's kind of lay out a few options
okay so we have um

behind the first door
we'll use letters to talk about them so
we'll have
i'll take the cash monty are you sure i
can reveal that one of the doors has
nothing behind it
okay so we're gonna have scenario a
which is
their um
r systems that are minimizing free
energy
so they're systems outside of that
circle so there may be systems that are
not minimizing free energy
but within the set
of living things
all of them are minimizing free energy
so let's think about what kind of world
that is
okay
so then we can have scenario b
where
there's living systems
there's non-living systems as well so we
could ask what puts us in one category
living or non-living but
there's living systems that are
minimizing free energy and there's ones
that are not i think we can say that
because
if we go with the weak claim
then
it's fair to say
that b is
ruled out
because this would say that there's
living systems that are not minimizing
free energy

so we can say like basically
weak claim
uh
uh disreputes
this area
the area of that crescent moon
that's being um sort of
invicted against
uh is distributes a word
disputes we'll go with that
the weak claim disputes this area of the
crescent it's saying well all living
systems have to be minimizing free
energy it's a necessity but it's not
sufficiency
so
then a third scenario
would be
like they're the exact same thing
so well
that would be sort of a
well you could have minimizing free
energy is a sufficient condition for
life
so that means that the life circle
is
we'll have to we can't show total
overlap but
the life circle
everything that's minimizing free energy
is alive
so here in a
there are free energy minimizing systems
that aren't alive
like a ball falling off of a cliff might
be minimizing its free energy just like
it's minimizing its potential energy or
a burning candle might be minimizing

free energy

but

it doesn't have to be alive so that's

these are free energy minimizing systems

in the crescent of a

that are not alive

in c

we have this

perfect relationship between the two

things

and that is a very strong claim which is

that if it's minimizing free energy

that's sufficient

for being alive

and if it's not minimizing free energy

it's not alive

so

this is sort of the the weakest claim is

on the left side

doesn't mean it's useless doesn't mean

it's inadequate but this is the weakest

claim this is the necessity claim

then we have the strong claim

here

uh

that

minimizing free energy is sufficient

well

this is

this is the strongest claim in c

that minimizing free energy is

sufficient for life

so anything that

you know if

doing x is sufficient for

greetings blue

hello

cool light

we're um in the slides on 24 just
talking about necessity and sufficiency
c is the strongest case
it's saying it's sufficient
and so you're either going to be
minimizing free energy and therefore
sufficient to be called alive
or you're doing neither
and then b is a little bit of a middle
case that's saying there's some living
systems that are not minimizing they're
free energy but some are
so these are three now are there other
possible layouts and then dean let's
return to your question about like
educating and how do we think about this
how do we distinguish these scenarios in
an education context
or welcome blue if you would like to
say hello or just what
makes you excited for this paper
um i loved this paper i thought it was
great um and i have been listening along
in the live stream so i'm kind of up to
speed with where you guys are and what
you were talking about but um
i yeah enjoyed axel's writing and um
i like like the fact that
i can
implement the free energy without having
to ascribe to pan psychism and also
without having to ascribe to the mind
life continuity hypothesis or all of
these things that we've been discussing
um like the necessity sufficiency
it's there like
i think that axel sets out a really
logical argument or a set of arguments

um
and
doesn't make any claims
that aren't um
justifiable with maths
yes
using a simple mathematical example
um we'll we'll keep this slide around
because we can wonder maybe is there
some well
another relationship that we could
speculate would be like
a kind of
something like this
like
life
there's systems that are minimizing free
energy or not
and then
there's systems that are alive or not
and sometimes they overlap but they
don't have to overlap but there's some
special living systems that are doing it
and there's some that aren't and there's
some that are neither so this is the
most
uh we'll just call this one
situation d
d is the most
is the most
flexible
scenario
because you could have
a or b i mean well i won't use those
letters you could have life free energy
minimizing
both or neither so d is the most
flexible case

c is the most restrictive case
and then we have two possible nesting
relationships where one is nested inside
of the other
and
it's kind of
interesting to wonder what kind of world
do each of these exist in
um well where is the mind bubble
yes let's
should be around the whole slide is it
is that the mind the the modeler is that
why there's that relational or
personal framing of the paper because
the mind is the
mapping
of
these different venn diagrams dean
and i think it's really interesting too
daniel that in real time
we now have a slide 23 and now a new
slide 24
and although the we now know the order
that they
they presented themselves to us as
could we have started with 24 and worked
our way back to 23.
that's that's when it gets really kind
of weird and twisted but
it's also what i think the paper is
speaking to if we're actually thinking
about this and as a as another example
of a post potentially post diction way
of viewing things historically now
yes so
um
23 which now has a little
logo on it for those who don't have the

slide deck at hand

23 was derived heavily from the paper
right it was quotes from the paper and
then just a visual representation now
that was not in the paper but it could
have been interesting
and then that's right through
the specifics of who joined when and
what we were thinking about when that's
what led to 24 but
maybe we'll add another slide before 23
and then they'll both be a different
number

but

it does

bring up an interesting question

i think we could

look at well first dave thanks for
writing in the chat do you want to read
what you wrote or would you like me to
read what you have in the youtube live
chat

yeah um could i expand on it just
slightly please what i would have posted
if i didn't have the big limit

title necessary versus sufficient

two of the original concepts of general
systems theory have recently as of 1975
been resuscitated there's an admirable
discussion by russell akov

the first of the two concepts is that
whereas causal argument considers
necessary and sufficient or double
implication conditions

systematic theory or cyber systems
theory or cybernetic argument deals
primarily with a logic of necessary
conditions

see fred singer's discussion of the
producer
and product relations
and in summerhof this is called goal
relations
and i'd add
this deals with intentional action where
there are goals
uh
there's an old painful discussion going
back to
way before darwin that the world exists
only as push
galileo wanted to absolutely banish
goals of any kind any value had to be
absolutely
driven from science
if you say for the purpose of is get out
you're not a scientist
uh i continue
this notion underlies the more
sophisticated treatment of goals and
intentions conceivably partly specified
intentions
fuzzy goals goals with
um ill-defined uh
criteria which you might or might not be
aware of i would
the other concept of about the same
importance is the presupposition of a
systematic universe there is a tacit
assumption that things objects and other
elementary entities are interdependent
rather than being isolated units which
is the assumption behind the majority of
sciences
further as a result of their
interdependence or with the same meaning

of the supposition that things objects
and so on are not really unitary
these entities form systems and it is
systems which may be observed and
manipulated
so the notion that there's one and only
one way to approach an argument one and
only one order among
real things that are scientifically
respectable utterly wrong
utterly wrong if you can't work your
argument in at least two different
directions you haven't thought about
what your argument is
you don't know the boundaries you don't
know what it is
you don't know what a jellyfish is
you think you know what a jellyfish is
but you don't know the difference
between the inside and the outside you
may end up in the inside
so it's almost like if there's only one
way
to run through the argument that's when
necessity and sufficiency are easily
confused so if somebody just
drives in one lane from the beginning to
the end and i made my case
and that's all i have to say
well then
you've
implicitly through your actions your
enacted behavior you've uh
acted consistently as if what you had to
say was sufficient for making your case
the prosecution rests
and also it was necessary you think i've
included everything necessary and

sufficient so that's actually where you
get and here's maybe where we should
duplicate the slide again
you get that's an overlap of necessity
and sufficiency but then
if you did two
parallel arguments just you did one from
start to finish and the other one from
start to finish
now you could say
well
both of them
are
in their own timeline they were both
necessary even if they have totally
disjoint information that was being used
and then what is the relationship of the
goals
there and that
is kind of like dean's multiple eyes
open giving us the perspective to look
at goals
when we have multiple realizations
because it does help us distinguish
necessity and sufficiency because we
literally see
well it could have been done a different
way
so it being done in way one
can't be the whole story because we just
saw it what done in way two
dean
i'm sorry dad
you dropped off on me for a second there
but
so i wanted to ask david a question
about
yeah dean kind of drops in and out

he was giving some funny descriptors of
his internet hardware earlier it was
pretty funny
start again
sorry um so i agree with with what david
was saying and there's a there's a book
called the origins of order project and
system
and what the author basically says is
that projects are easy to
get trapped in goal directed behavior
whereas where if you
can also hold up the system
you can see
the number the sort of the multitudes of
different directions all at once and so
i was wondering
is is that
does that sort of fit in with what david
now was trying to speak to
[Music]
try to find out the author's name here
um
well stewart kaufman uh who is i think
he has become a little more active
recently i know they're pushing his book
origins of order really hard that's
something i read in like the 80s who's
going
this is a new one this is paul khan
okay
yeah okay k-a-h-n yeah k-h-n
and he's and he's basically trying to
use the argument that you can look at
the constitution or you can look at the
actual laws and and you hold up both
one is goal directed the project piece
whereas the other one is is is trying to

work from within and around
so that's it's that sort of between and
inside argument and that's not the same
as

as projecting yourself to a particular
outcome

yeah i can't i can't answer that right
now but i do want to tell a little quick
story uh when i was working as a senior
programmer analyst i was organizing this
project it was going to take a couple of
months

and my super is kind of looking over my
shoulder and i have the criteria and i
have a big flow chart and he says oh so
dave this is how you're going to do it
and i said i haven't decided he said
well wait a minute what's missing i mean
you've you've got the whole thing laid
out it looks like it ought to work and
says yeah i haven't decided i've only
done it one way there's no decisions
involved in that

i've just stated the requirements i
didn't decide how to do it so then he
went out and had a smoke
i ended up using a much better technique
that saved us a huge amount of rewrite
that nobody would have noticed because
they would have said oh yeah the answer
go for it

it's almost it's the zero to one which
we've talked about and it's also related
to

act

infer serve

infer you'll have something to infer
about

when you've acted whether getting that
first draft out or just doing a private
voice memo or
whatever stigma g you need you'll have
something interesting to infer if you
act i promise
and then maybe you'll be closer to
service than you expect
but infer
is still within the realm of like well
what would be sufficient to get the job
done
but if you go well what would be
necessary to do this one way
this is what would be necessary we could
um walk
from california to new york
that would be necessary for getting to
new york okay
it's gonna take too long i just don't
have that kind of time
what would be another way to do it
so there is a almost
a necessity and a sufficiency
in
action and inference
where action has a lot more to do
what's needed and
inference or thought maybe moves us more
towards thinking about
what's sufficient
not sure about that but also thanks for
sharing this um
this resource
well and back to what david was saying i
in the work that i was doing i often had
people who
who are very well versed in setting up

all kinds of sophisticated projects
asking me where's the project in this
and me having to jump up and down and
wave hands and say this isn't a project
that's why
we're doing it we don't know how it's
going to turn out until we do and then
we can look back on it and say well
did we minimize free energy we weren't
using minimizing free energy but
because i wasn't you know first in that
lexicon yet but that's essentially what
we
were doing
we wouldn't know until we'd arrived that
we'd arrived
and that's that's really uncomfortable
for people who were constantly saying
show me the milestones
okay well i can show you the stages
and i can show you how identity identity
is now in turning into identifying as
people are are constantly processing who
they are as they're
on board this
this
this idea but i can't tell you
what it's like because it's not
necessarily goal directed other than the
fact that we want to try to keep it open
as long as we can
and figure out what logic people are
using in order to be able to untangle
and
and pull things out of hats that that
others don't know how to do
so yeah
one other thing that makes me think

about is grants applications
so a grant application might frame
things in terms implicitly
or explicitly in terms of sufficiency if
you fund me we're going to
fix this disease like this is the grant
that will be sufficient for fixing this
disease
whereas saying that it's a necessary
step like we do need to understand how
protein one two three works
it'll be necessary but it's not the
whole picture it's not sufficient
so so much
when we really turn it over
is related to
necessity
because there's so many things that are
necessary
and so little
save for totally exhaustive examinations
of really contrived systems
so rarely do we find sufficiency because
if you can define sufficiency
you can make that bacteria in the test
tube de novo
and then life
it's kind of like puts a little twist on
it like what would be necessary and
sufficient for having a bacteria in a
test tube
putting a pre-existing bacteria in and
having it replicate
so life as this paper explores
it sort of threads the needle of
necessity and sufficiency
because the past
is necessary and sufficient for living

systems to be in the state that they're
in now

and so that's like the generalization as
well as the minute particulars

if anyone wants to ask a question

otherwise i think it's good to take a
second look at

the numerical example yes blue first

so just like

what got us to this point the past is
necessary but not sufficient but it just

just echoes the title like how did we

even get here right

yes

so let's look at that

bacteria and then of course we can jump
around

anywhere else

okay so the numerical example is of a
sensing

and feeling a sentient bacteria not
necessarily experiencing

and action isn't even in the loop here

but it's a sensing bacteria

and so

the organism

is inferring its prior beliefs about the
cause of the observations it makes

so it has a prior

let's just go over the um sort of

letters again so there's a class of

events for the organism so this is like

we need a special color or some metadata

tag like how can we tag map

and how can we tag territory let's just

use like

one set of colors or one font like map

comic sans

territory some
calligraphy script or something like
that
for the organism
of the organism now it's our map of the
organisms map
a and b
are events that are part of the class r
and so r is like the receptor state
and then
a and b
are inferred downstream of a chemical
signal
which is s so s is like for signal and s
is whether the actual molecule on the
outside is the a or the b
so this is kind of a discrete binary
choice
it's briefly discussed in the paper like
what happens if there's multiple
molecules or one could imagine that
what's being inferred is a continuous
variable like blood sugar
but alpha and beta
are s that's the outside stuff
and then a and b
are the organisms
receptor inference
blue
go ahead and then i i'll just both end
yeah so just to sort of
look at it in figure one or in the
formalism uh block that's in the paper
the organism has a prior and it's the
specification of priors
that differentiate
bayesian statistics versus frequentist
statistics

among a few other pieces but that's really the key piece and so the prior p of r is 80/20 in favor of a so a priori before how did you get there but a priori it's 80 likely that it's a and 20 b before any info comes in then there's a likelihood mapping so that's the the distribution of the signal conditioned on the receptor state so we're talking about what's the probability of α outside conditioned on a and b and so it's 70 a and 30 for b and vice versa so what's interesting about this is that's a pretty noisy signal it's really only saying it's like if it were 50/50 it'd be a useless signal because it wouldn't knowing it one way well it would also depend a little bit on the frequency of α and β but this is a very noisy signal like 30% of the time the receptor is giving the wrong answer so this is a way to show that with multiple uncertainties connected the right way you can actually tighten your estimate because in this example with the 80 prior on a and then a 70

likelihood mapping
which is also a that's more like classic
active
jargon
after observing alpha
it turns out
that the estimate of a
the estimate of alpha
the estimate of state being a
conditioned on alpha being observed
tightens that prior
from 0.8 to 0.9
so
there's still a tightening of the prior
by a significant amount
even though
the mapping between the molecule and the
receptor state
is somewhat noisy
so that's like a bayesian bacteria it's
just applying bayes rule
to this
simple binary
outcome that it's doing inference on
blue any thoughts there
yeah so uh you know i this goes back to
something that dean brought up in the
dot zero like is oh and it's actually
something that axel wrote in the paper
like is the prior subjective or
objective
um and like the mis dysfunctional
requires and the survivability of
organisms with dysfunctional fires
so i just feel like i mean i know axel
says it's subjective but i feel like
it's nested so i feel like there's the
prior must be a subjective prior over

and objective frequency right like so so
if i frequently observe you know
sunshine outside when it's warm or
something like that like if mostly when
it's more when i feel warm i absorb
sunshine so when i have these two things
linked like based on that frequency i
then make a subjective assumption and
and that's how i feel that the the
structure of the prior um
kind of has to be
otherwise like it doesn't make sense to
put the math into the prior
um like like in my mind like if you
can't
if an objective prior is based on
frequencies but a subjective prior is
not a subjective priors
i guess but it's a guess over the
frequencies i don't know i feel like
there's like an extra layer that needs
to be tucked into the subjective prior
if you're gonna put math there
one
way that that's dealt with empirically
and then dean is what's called
parametric empirical base
and in parametric empirical bays you
kind of start with just
the prior being what you observed in the
past so if you had eight sunny days and
two rainy days
that's where that point eight and point
two prior comes from
so
it's um
not a way around specifying a prior it
just starts your prior in a place that's

empirically grounded
dean
yeah um
so i think in the point zero two i was i
was
trying to figure out how
the likelihood is really
the
the kind of background that we have
and that's what blue is kind of talking
to you and i was wondering about in
terms of context because if i see a
black dot on a on a white background i
think of period whereas if i see
a white dot and a
in a
in in black background with a whole
bunch of white dots i think how many
light years away is that concentration
of hydrogen right
so again we could d we could
decontextualize this
but
but if we do
d decontextualize and this is where it
gets really hard for me i get lost in
all of the terms but
if we do
decontextualize it
does that
does that help us in any way understand
both the subjective
and
objective factors in play that's all i'm
i'm wondering about and again
i get i get lost in the number of
different ways to be able to describe
this that's part of my problem i have an

upper limit on how much of this stuff i
can do once if you want me to focus on
days
i can do that that's hard enough for me
to concentrate on that
but if you actually think about this
okay so what does this actually imply in
terms of entailment
then it gets really really tricky
because i can't just focus on one thing
at once i gotta sort of think about what
all the implications of that are and
man can never go off in different
directions all at the same time so i
don't want to be too tangential here but
yeah that i i'm still not sure whether
the subjectivity in isolation can work
i'm not saying it can't i don't know
thanks dean blue then dave
so in like the empirical parametric base
that would be like an objective prior
like do you have subjectivity in
empirical parametric base because like
that's what axel was describing in the
paper you do you can have an objective
prior based on frequencies like it's 80
of the time you know sunny and 20 of the
time rainy and so you you get these
frequencies and that's what goes into
constructing an objective prior but but
when it comes to like an organism and a
choice or even like a scientist
conducting an experiment you know like
why would you even do the experiment in
the first place you have a guess right
like you have something that you think
will be the outcome and so like a
totally subjective prior is like well i

think

sixty percent of the time it's going to

be

blah blah blah and like that's a

completely subjective prior it's just

your best guess which which if you're

gonna try to really do objective

bayesian analysis

like you need to have a flat prior not

your best guess right you need to have

your prior needs to come from previous

observations and not you guessing but

but you could very well like not you

know do a very empirical or objective

bayesian analysis and just say my guess

is this and then you count make the

observations observe the frequencies you

know you go through the entire time step

and then at your next time step there is

that tightening between the likelihood

the previous likelihood and the prior so

that that is um

i mean as far as i understand it now i'm

definitely not an expert but so i think

like the subjectivity and objectivity

is

i don't know i think that there's some

some link there like i get it for

scientific bayesian data analysis like

you want to have a flat prior you want

to do objective analysis you want to use

you know the previous frequencies or

whatever but

i think that um you know in terms of a

mental calculation

it's also over frequencies or it's your

best guess like what do you think will

be the outcome and it's not always

doesn't always line up right

thanks blue dave

yeah um

the term

decontextualized i i kind of that

shocked me when professor jaron started

using that it's just he contextualized

what that that's just some some

jerk opens his mouth and blood he

contextualized his feelings at me and

you know if i want to know what's going

on i hyper contextualize i've been

particularly attentive to my own

prejudices and my own viewpoint you'll

think about what goes you know einstein

hated the word relativity he wanted to

come up with something like hyper

contextualized or the real world or the

the um

the tensor description uh

where you don't have to ask where are

you because you have the structure of

the world

um

what do people mean by by d contextual

odds uh is it is it a term of praise as

well as a term of contempt or have i

just always misunderstood

blue

so i love this term hyper contextualized

and

i i don't know that i would use it in

the way that you use it dave i i would

use it like

um

i would play out all of the potential

like what ifs like

like let's talk about counter factuals

can we can we put it into you know how
many different contexts will it fit um
and that's when when i think about hyper
contextualization it's like playing out
all the potential scenarios for me
thanks dean

yeah and building building on what blue
just said

um

so i i saw a youtube video where a guy
figured out a way to
make a target move so that no matter
where you threw the dart as long as you
were directing it towards the dart board
the target would create
the algorithm would create the target
moving so that you hit a bullseye every
time

and i was a phys ed teacher

i was like oh this is amazing and i took
it to one of my colleagues

who shall remain nameless and i said we
should show this video

to the kids and he was like why

and i said well think about it every
time a person moves their hands
to catch the ball

they're making the same kind of
adjustment and it was like a light went
off

a lot of the things that i was that i
was coaching were then decontextualized
and the proof was in the fact that when
you saw

something out of context that was
running the same parallel function
you didn't necessarily recognize it so
this was a really i i love hyper

contextualized

so i wonder if that's like abstraction

or

d

irrelevant contextualizing you take away

the irrelevant part of that

setting which is oh it's hands or some

other specific system

and then that

paradoxically

gets you to hyper contextualize because

it does

highlight the parts of the context

that matter dave

and there's that famous phrase from 1923

consciousness

of abstracting

training people to think scientifically

one of the first things you do is get

them really aware that they're how much

they're missing you're abstracting away

from at least 90 of whatever you're

looking at

uh and since i've got the mic um

there was the question of how a fielder

catches a baseball

a fly ball here it comes it's coming out

of the sky at you

and you know the good fielder knows how

to do that you ask him how you well i

just i just go to where the ball is

going to be

but he can't tell you how and there were

a bunch of people mathematicians and

programmers and physiologists

and philosophers trying to how in the

world are they doing this it looks like

they can't because we can't come up with

an algorithm that doesn't and they
finally
watch them
and i guess they got one one one guy
that's high on
whatever it is openness the big five
uh and and they says well what are you
doing how do you know that you're gonna
catch the ball oh i i know i'm gonna
catch the ball when i look at it and it
doesn't move
it doesn't move yeah it just hangs there
and i put up my my mitt and it since
it's i know where it's going it's going
to my mitt
and it's not moving and it's in my mid
and it didn't move then either
that's how they get there they figure
out what point in the field makes it
look like the ball's not going to move
as it comes toward there and that's
where you go
and then just
hangs in the field till it's in your mid
classic baseball physics
the thing i thought was have you ever
been high on openness
but then it's true there is a simple
algorithm like if you're looking and
it's to the left then you should go to
the left if you're looking it's to the
right and then you're
cool active baseball and anticipation
and the path of least action
blue
you want to talk about high on openness
i don't know that it reminds me of our
conversations with adam saffron right

and so

like what the psychedelic like

i would like to see like the psychedelic

like comparison with big five like the

integration of those two kind of things

in terms of active inference because i

do think that um the openness like

increases a lot under under the

psychedelic influence and so it's um

not have you been high on openness but

has being high ever made you more open

maybe is a different question

and well a few things there's sort of

the um rebus sevis albus relaxation of

beliefs under psychedelic strengthening

or just the altering of beliefs with

fristen carhar harris saffron

and what is being what priors are being

opened which priors are being tightened

and that was sort of at the heart of the

rebus versus cebus and then the sort of

um hegelian synthesis in albus and then

also in terms of how we got here

the first two weeks of january was live

stream number 13

with

adam

and colleague

on the cybernetic big five

and so that actually sort of started our

year

with thinking about how

frameworks that

a priori have nothing to do with actinf

or fep like big five was a principal

component analysis based

variance decom uh various partitioning

on psychometric data

yet
we saw how
active could help us
reimagine that
and generalize it
while also bringing action into the loop
so then it's kind of cool to see how the
personality traits come back into play
uh dave
and then we're going to return to base
and then we're going to move into the
variational free energy
well stop me if i already mentioned this
but um
there's been some
uh
criminological studies of the way
psychedelics are used by
as a recruiting tool of violent cults
you go to a rally
in favor i will not say whom
and if somebody seems like they're
really grooving and really into it you
give them a little ecstasy and that
locks them into the mentality
and if he's kind of getting fired up but
but he's not quite with it you take him
home give him some lsd and take him to
the session
and that moves him out of what he was in
and moves him into where he needs to be
in order to do the right things in life
maybe another a little bit of ecstasy
now and then over subsequent months to
just get him locked back in where he
needs to be just like
just like safran was saying you either
get him out of the of where he is or you

get him right back right deep into where
he is
well there's a long history of
people engaging in experiences taking
substances to move their
psychodynamics one way or the other um
i think
the active streams verge on it but we
also
remind people to be responsible and
legal
with their behavior and with the one and
only brain and mind that they have
let's return to the bayesian maths and
then see how that bayesian bacteria
moves into the variational free energy
bacteria
but i pulled out that one
quote from a um online statistics
resource so we can you know uh
the 0.0
is the frequentest bacteria
and so the frequentist
is basically looking at the outcome of
an experiment
you flip the coin ten times it comes up
seven heads and three
tails now that's not the most likely
possible outcome
for a fair coin but it's totally not an
unexpected outcome
so
in that case the frequencies can produce
a 95
confidence interval
given the variance granted by the sample
size
and say okay if i get um within these

two numbers

um then i'm going to fail to reject the hypothesis that it's fair if i'm outside of those two numbers with my empirical results i will reject at a given p-value a given alpha

and so for the frequentist there's no sense in asking about the probability that the coin is fair it is either fair or not fair so that's like accept the hypothesis

implicitly because of the failure to reject or reject the null hypothesis in favor of a specified alternative the frequentists make statements about the probability of a sample after making an assumption about the population parameter which is just the kind of average out approach average out value of the probability of tossing a heads the bayesian

in contrast

starts with information about coins so specifies some distribution so it could be like a bell curve

a gaussian centered at 0.5

it could be a super tight prior could be a very loose prior and then the extreme case of the gaussian with infinite variance is the uniform distribution and so that is like a flat across all the possible values

that probability assessment of the heads proportion is called the prior probability each person might have their individual assessment based on their personal experience called a subjective prior

alternatively a prior probability distribution might be selected on good properties of the resulting estimates called an objective prior

i don't know if this is the only way to use subjective and objective in the bayesian context but this is just an important difference between frequentism and bayesian so

here is our bayesian bacteria again it has some a matrix that's mapping the signal conditioned on the receptor state and then it has a prior over those receptor states that bayesian bacteria

then again it's a bayesian perceiving bacteria we're not bringing action into the loop yet but we're just assuming like if you had the right estimate then it would be easier to know what to do correctly

um this is called exact base because the exact numbers are plugged in and then there's some exact number that comes out and it's pretty much just granted by bayes rule

what's the issue

well it's unclear whether living systems have sufficient computational power to accomplish that especially when there's just massive multivariate landscapes

so

what is the answer in the fep

well

instead of doing exact base

there's what's called variational

bayesian inference

so here we're going to propose this

q

function

and the q function is going to be from a

family

that is specified a priori

and it is more easily fit

than just sort of the exact base

and then the way that we compare

is not by comparing at the end the

posterior likelihood

but rather we can compare

different

options

by comparing their

free energy

and this is how free energy is being

defined here

it's the negative

sum

over r those are the receptor states

a and b

of some q distribution

and then the natural log of the joint

distribution

of the states r and the signal alpha

multiplied

uh then

divided by q_r

and so then in dot zero we explored a

little bit about how that's like the

recognition model that's like we're

going contingent on alpha and now we're

doing a hyper parameter estimate like

alpha we're going to a

and then what we're not going to be

exploring here is like a generative

model and then we talked about how those
map onto perception and action in the
world

so then the numerical example shows that
okay so the bayesian bacteria it did the
right thing it didn't go to that rally
that dave was telling it to go to
but we can also have a free energy
minimizing

creature bacterium

and we're going to see a case in which
the free energy minimization does the
right thing and we're going to see a
case where the free energy minimization
does the wrong thing

the difference between these two cases
is going to be the prior

that's set

and so that is the crux of the argument
that minimizing free energy is not
sufficient to be alive aka make the
right decisions

because both of these examples
are going to minimize free energy and
one of them is going to minimize free
energy in the right direction

and make the right choice because it had
appropriate priors a maladaptive priors
can minimize free energy but still
result in the wrong outcome

so in this case

we have the free energy calculation
for

um this

80 20

prior with a favor of a

and

contingent on observing alpha

the free energy of this is 0.47 which
doesn't have an interpretation as a
probability importantly
so that we're not within that zero to
one bounded space
and then
um in contrast
the representation of b
so that q r setting that to b
that has a free energy of 2.2
so free energy minimization
saves the day in that context
however it's not simply because the free
energy was minimized that the right
decision was reached here is a creature
with a prior on b of 0.8
and then
we see that
just barely
upon observing alpha
the free energy
of representing b after sensing a is
0.96 and the f when representing a after
sensing alpha is 1.094
so
the ensuing posterior likelihood
is 63 percent in favor of b
and 37 in favor of a
so
we sort of
did move the needle towards
a
because we observed alpha
but our prior was so far off
that just one cycle of free energy
minimization still results in the wrong
outcome
that's the crux of the numerical example

provided in this paper
and the only point that it's being made
not this is how you'd model a bacteria
with the fep
the point it's making is free energy
minimization is not sufficient for life
dean
so daniel i got a question to ask so
i don't know because we're just talking
about sensing we're not talking about
you mentioned
doing but i is it is it is it doing or
is it directing so
because of the sample
sense
and because of what the
math
points to should we think of this kind
of analogous to a compass we're pointed
in the right direction we still haven't
actually acted or developed a feeling
from the sense yet we're just we're
still pre
doing we're we're we're orienting is
that
or am i
wrong
so if anyone else has a thought after
this so let's let's i think there's two
few aspects to this compass and
orienting approach
so
this little toy motif
about the recognition model
about having a prior
overstates in the world and then getting
some input
that could be plugged into an action

model

in a slightly more elaborate case so for
example as written in the caption for
figure one

more biologically realistic descriptions
of behavior require discussion of active
inference

they require it it's necessary and

sufficient you just have to do it

behavior is the result of a different

inference process that of an action

policy so π

is not here in this bacteria

this involves more priors namely about

the transition between hidden states and

often about preferred sensory outcomes

so inactive inference with a partially

observable markov decision process we've

seen how there's like a matrix of the

transition frequency b

between the states at one time and the

states at the next time

and then policies are inferred

based upon which policies minimize free

energy

which ones are going to have the best

chance through time

of bringing states

into alignment with preference

the other thing that it made me think

about with the compass

is like the exact base

just

returns u_m a probability

and that is sort of like one compass

needle

uh

giving you just saying okay you're going

north by northwest or whatever it happens to be this is your one estimate you could be wildly off but given where you were and then just one blip of information that's where your compass needle is now pointing to free energy it's almost like there's two compass needles in this binary choice and then we're going to select which one has a lower free energy because we're looking the fact that this is 0.478 f and this was 2.27 they don't have a natural interpretation just like entropies you can talk about the difference in entropy but something having like four doesn't have an interpretation in the world and so it's kind of um maybe like there's two different compass needles and we're looking at the difference between them or we're going to select the one that has a lower free energy dean yeah i don't want to say that orienting isn't a form of doing as well it's just that what the what this shows what this shows me is that like you i you i cut out again but when we're when we're trying to when we're trying to determine a and b with a background um we haven't taken an action step we

haven't developed a feeling yet
do we do we categorize that's
differently than if we were to look at
this through an active inference lens
where
we would then
push a policy forward right
so i'm not sure it is the orienting
fitting you fit into the policy piece or
is it pre-policy that's what i'm trying
to get clear in my mind
good
question
it makes me think about the uda observe
orient
decide act loop and about our
recent paper that connected active to
the uda stages
orienting
i think in this reduced example
the um the north pole north star
whatever it is is the prior
like that is what is oriented
relative to
so that's a purely recognition model
sense of orienting whereas in the
physical
realized world
orienting has
much to do with action and wayfinding
but this is sort of in the recognition
toy
there is still an orientation
of the prior
but i'm not 100 sure what do you think
dean
what i what i find interesting is is
that when i look at this

and i see a like a 0.8 and a 0.2 it's
very clear that we're somewhere between
0 and 1.
but when we're actually getting to that
wave finding space
and we're talking about entailment
one whole over two wholes
is different than two over one
so as soon as we move past one
into some of the more as you say
material
realities that we have to face down
that's when those actions
become
material like we can see what the
process ontology is but what
axel did was he brought the process
ontology down to it even
even closer to zero point because he was
using free energy
principle and he's talking about free
energy principle and bayesian stuff so
again sometimes
i have a general sense of this stuff and
i
just enough to get me into trouble and
be an insurgent and
and
point it people who are really really
smart and kind of go i think i know what
you're on to but it doesn't get me deep
enough to
to be able to write a parallel paper to
axles and sort of
either build them up or or tear them
down so
yeah i don't know i'm just that's why
i'm asking i don't know where oriente or

orienting

fits in this because i think it fits in

the zero to one space and i think it

fits into the one to three space

as we as the dimensionality blows out

that's back to the subjective

objective

crossing back and forth and again

i come off sounded like a complete idiot

here because there's somebody way

smarter than me who can blow my argument

up but at least i'm putting it out there

i'm showing up

oh you're you're

by being on a live stream and talking

about it you're like 80 of the way to

the paper

um

dave

is there something in this context that

would correspond to urgency

and i'm particularly thinking in

the terms that freud

was concentrating on when he talked

about how to get people to

be less neurotic he says

if you're

the the point that a person is in real

trouble as a neurotic is when the

dominant

factor in his life is reducing his

anxiety

and what we want to do is to get him to

tolerate his anxiety tolerate for

instance ambiguity not knowing is my

wife mad at me or not but don't get

crazy

sometimes you just can't do anything

about it

calm down tolerate the anxiety is there
something and i i'm just taking that as
a case of urgency urgency seems like
something that would fit into this
literature better

blue

so i just want to um like so i don't
know about urgency i'm going to kind of
derail off of that and more hook onto
this toleration of anxiety

or toleration of circumstances that are
not preferable for like a later more
preferable outcome

and so this is something that is taught
like i don't know in self-development
like a lot right like you have to put
yourself out of your comfort zone but i
mean also like

in like the meditative traditions like
to learning to like be with that
discomforting feeling and not like you
don't have to like get all upset and do
anything about it um so i i just wonder
if that fits like

it's a different different thing
different than urgency but what about
like where's the temporal

um

there has to be like a loop right like
uh i'm gonna tolerate this bad thing for
something that i think will be like a
more predictive outcome later like
how can i bathe that
that would be like an interesting
mathematical um

thing to try to work out

cool i added a uh slide

here here's the eisenhower matrix

so

um

the x-axis is urgent to not urgent and

then the y-axis is important to not

important so

one of the reasons why this is such a

useful and well-known tool is it

separates urgency from importance

because there's things that are urgent

like a chat message

that urge you to do something or they

have sort of a short time horizon but

then there's things that are important

or not important

and then

there's an orientation

when you're figuring out what is in

what quadrant kind of like the rumfeld

matrix as well

and then that

has implications for action

so let's see

urgency and anxiety dave you mentioned i

don't know about the freudian urgency

concept much but it's of course with uh

psalms with fristen and the project for

scientific

psychiatry there's some clear links

there so it wouldn't be unexpected

and then

the idea that the imperative

is to reduce our uncertainty through

updating our model learning and through

action

so the reduction of anxiety is in

imperative

and then sometimes it crosses a

threshold towards being truly urgent
so it's kind of always in the background
but then other times it really is urgent
but it's always important
and then blue i
agree
delayed gratification
it's like
having the time horizon depth
to take the right policy selection today
so that your expected free energy
is higher in the future so it's like
would you rather have one dollar every
day
or
a thousand dollars in one year
depending on how badly you need the
money today and what you think you'll do
with it in the meantime and all these
other things
one person given their priors
might minimize free energy and come to
two different conclusions
and i think that's where we return to
this subjective versus objective
fep take
which is like
a person could
rationally evaluate
given their subjective priors and choose
to take that marshmallow
maybe that is just the waiting that they
have on their priors
or maybe somebody does the opposite and
that is just where their priors and that
stimuli
that's the action selection that they
take

and so
um
not sure how all these pieces tie
together
but
i think it
it
speaks to the
the the logic of action
and inference
and we want to have a logic of action
and inference or a framework for it
that is
broad enough to accommodate like making
bad decisions
if our action and inference framework is
only
tightly constrained scenarios that are
addressable
then it doesn't really help us as much
with real world scenarios
but it will be very cool to see how a
lot of these
daily
phenomena
connect
to the things that we're talking about
here
so let's see what else we had in the
pieces and and um also if if axel is
listening or wants to join for a dot 2
next week or for a future.³ i think
there's probably many things we could
ask and
what brought you to
write this paper and then
where do you go after this one
but pretty much the paper

delivers the
mic drop the coup de grace
with the free energy minimizing bacteria
which has bad priors
or let's just say maladaptive priors and
it minimizes free energy and makes the
wrong decisions so that's where that
section concludes
then there's a jump back to the more
philosophical framing
so although free energy minimization is
not sufficient for life
there might be an entailment relation
that goes the other way around
if you are alive it might very well be
because you did something like
minimizing free energy
that entailment relation is that which
corresponds to the weak version of the
life
free energy entailment relation so
remember way back when when we were
looking at the relationship between life
and minimizing free energy
this example has
decimated the strong claim blown out
make a youtube reaction video
um
but the weak claim is actually supported
because it seems like making adaptive
decisions
is consistent with free energy
minimization just like it would be with
making adaptive bayesian decisions
but we can say that it's not sufficient
to do so
and then the name of the section free
energy on a wing and a prior

so

yuck yuck yuck anyone can give a comment

on that but we just sort of

were reminded of two winged creatures

that we have talked about only a few

weeks ago in 32

there's the winged snowflake and then

the lepidoptera to a flame the moth to a

flame over here

and about how

in this snowflake example so go to 32 to

check it out in more detail

but it's like

the snowflakes with wings that act

adaptively

are going to persist and those are the

ones that are going to be

looking as if they've minimized

free energy in selecting the correct

action policies

whereas the ones that don't exist just

aren't if it makes bad decisions it's

going to fail to exist

um

so

it's an interesting

section

and it uses sort of evolutionary logic

in a little bit of a different way

because the evolutionary answer for

everything is just sort of like that's

what was favored by evolution

selection

uh

why does

this organ have this shape it's like

well that's what was selected on

and so that's just that's just why it is

that way even though we've talked about
tim bergen and other things we know
there's multiple y's
but here
we can now see the adaptivity of priors
instead of relative fitness
merely
blue as we know more information here
soon
but the the adaptivity of the priors
will be looking
as if it has been minimizing free energy
at the population level
so whether we frame natural selection as
increasing individuals with fitness or
increasing the mean fitness of the
population
but not necessarily the relative fitness
we can also do the same
but rather with a malthusian exponent
or
some other kind of
economic
fitness parameter we can think about it
in terms of the free energy minimization
dean
yeah i keep falling off here so i'm
gonna i'm gonna
have to get past all the hidden stuff
that you guys talked about when i wasn't
uh staying on on the broadcast but
what this points to me is interesting is
because
back to the previous slide you talked
about my fur
rip dean
please f in chat for dean
he'll rejoin in just a few seconds

sorry guys continue

yeah is it a wing my first question is

is it a wing as wing

and then second what but does it still

serve a function like could i use it to

fly

and so that's where the adaptive piece

actually comes in for me so again

there's two questions necessity and

sufficiency right and and you can even

see it in the representations is there

enough information in the representation

that i identify

as wing and then the second part of it

is

well then

i can see it can carry a moth or i can

see it can carry a snowflake that's the

function that it serves and so that to

me

again is the minimum of two and making

make sure that we don't

blend those two things up because it's

the fact that we hold both of them as

discrete and each serving a different

purpose that helps i'm gonna i'm gonna

jump off prematurely because i tired of

having to catch up and slow you guys

down but i'll watch the video and

seeing the point too thank you peace

dean

all right

okay

cool

so we walk through

much of the paper it's not a very

long paper

and it has just one core numerical

example again with a bayesian walk
through
and then two variational bays
one variational base that basically
recapitulates the first bayesian example
of an adaptive prior and then a second
counter example that provides a
maladaptive prior
that is connected to
evolution
and to the idea that we're only going to
see adaptive priors moving forward
because if that
model of recognition or that action
model is maladaptive
it will be seen less and less as time
goes on blue
so something that axel didn't mention in
the paper or maybe um didn't mention or
if i if he did mention it i didn't catch
it but
so the
adaptive priors always fit a certain
niche
right and so like that you're only
adaptive you know if you're like a
thermophilic bacteria right like you're
functional in a hot spring maybe not in
like an ice patch right so i think that
there's
you know if you if you take the
thermophilic bacteria out of the
hot environment and put it somewhere
cold then it's it's maladaptive its
priors are maladaptive but
only because it's just not fit to the
the correct niche environment and so i
think that there's this

you know the importance of of the niche
environment and also it makes me wonder
a lot in terms of evolutionary like
conceptually like
you know what if

it's always we always think about like
the best fit to the niche but but what
if it's like

the best generalist like is there like
the evolutionary i mean i guess that's
like the cockroach right like like the
evolutionary generalists that can go
anywhere be anywhere survive in any
niche survive on anything like is that
the best like how do how

how does that like

like a prior that fits no niche or fits
all niches how how does that like play
into this evolutionary paradigm i mean i
guess the cockroach will prevail
good question maybe this is related to
our discussion about context
uh having the prior of 80 20
and then we observed

um

[Music]

alpha but we could have flipped that if
it was all of a sudden dipped into the
beta ocean

then that prior

would be as wrong

as the maladaptive prior was in this
paper

so

whether or not it is adaptive as you
point out it is absolutely about the
statistical regularities of the niche
and that's the fitness of the model

fitness of the organism to
the
generative process is the generative
model
going to be fit to the generative
process and then generative processes
change things change out there in the
world
dave
yeah if the uh the work on my medics
going back to dawkins and the the
english lady that did
a lot more development of that notion uh
if that can be adapted to free energy
that will give a lot of very detailed
discussion of generation after
generation of the co-evolution of
adaptability and that
which is adapting
the the notion that the brain has been
adapted to imitate and to remember and
to pass on and to
recast concepts into many different
modalities
so do what blue was asking
yes
agreed with that
we can then just maybe look at the last
few pieces in the future directions
and then i think
see if we have any ideas for things to
discuss and point to and then other than
that just invite everybody who's
interested in listening
to join us for point two because we've
covered
much in point one in zero
so it's our last

journal discussion of 2021

come join

and

share how you got

here

and then what action selections you're
going to take in 22.

the last section is

free energy minimization as a historical
scientific principle

so that's a very fascinating idea

because

it's about anticipatory systems

yet it's a historical principle

that's where we get into the difference

between prediction and post-diction

postdictive scientific statements are

concerned with what must have been the
case instead of what will be the case

a statement such as the minimization of
free energy may be necessary if not
sufficient characteristic of

evolutionary systems which was from the
first and staff in 2007

is probably such a postdictive statement

and then axel claims that that statement

should be interpreted according to the
weak claim rather than according to the
strong claim

so interesting that even back then

the necessity insufficiency was being

used and arguably used properly

so then

what happened

how did

we get off the rails

or how did the strong claim come to be

and

the paper

i think

makes a powerful argument for the fep as
a postdictive model

but that doesn't mean that it can't be
used in an anticipatory way for
anticipatory systems moving forward
and

doing inference on the consequences of
our action which are also in the future
so how the uh

past and the present and the future
are linked

that's more of a dot too

how are the historical sciences
anticipatory sciences
and uh

i don't know is there now sciences

no that's what i call science 21

what else

would people like to discuss in dot one
blue

so

i would like to maybe get into more
about the living versus non-living and
maybe talk about um

you know this this idea of maintaining
structural integrity like is that a
quality of life or do things always
maintain structural integrity and like
does the fep apply when structural
integrity is maintained

and there's not life

i mean it's like i mean we talk about
like free energy minimization and
chemistry all the time like you form the
product that minimizes the free energy
and so i know that there's some

differences or some
a lot of extrapolation that has to be
done to get from thermodynamics to the
fep or
not extrapolation but like
it's retrofit
is maybe the right word um so i don't
know i would like to maybe see some of
that
where does the fep apply in situations
that are not animated is it
computationally is it relevant like when
we're talking about robotics
um

[Music]

yeah
that type of thing
yeah and is animate the same as sentient
like animate sentient
inanimate
not moving not animated
in sentient
like

can we just have an active model
that doesn't have any affordances or
where it only has one affordance
so it's still doing a recognition model
but then it's
policy selection is extremely
constrained
or like

so i mean i i don't even like to like
use life and non-life because there are
some things that like blur that boundary
also like it gets into like well what is
a virus is a virus alive like
uh and a virus is very exploitative like
it has dna like there's

some things some qualities of viruses
that make me feel like they're alive but
they're technically not
so
like what about if we just say dna
containing but then it's like it's this
test tube full of dna alive like i don't
know
it's hard
and so instead of even sentient
inanimate like alive and not alive
i don't know
yeah i i think the the the paper's
succinctness and clarity of argument
hides a big
question which is okay
so if it's alive
who told you that
so
if we can't go from
minimizing free energy to inferring
what's alive
then how did we get to here
right or what is even alive mean like i
don't think it's fully defined like if
we can't fully define death
like
i mean how do we fully define life right
so i don't know it's this there's this
big sticky
i don't know weird area i mean we kind
of instinctually know what is life and
what is
what is alive and what is not alive
um
anyway but the
i mean are the
crystals in bone alive

well they're part of a bigger system
that people usually don't have a problem
saying is alive but then why should what
people think matter

like

especially if we believe in evolution
and we think that our cognition is
oriented

not necessarily towards like ontological
realism

then why should it matter what one
percent or 99 or even 100 of people
agree upon

or even if we zoom like way out and get
super meta like how does life maintain
the structural integrity of life

that's an interesting one to think about
like like because i mean it's dependent
upon the niche environment which
includes many non-alive things like the
crystals that are inside of bone

um or inside the inner ear which help us
maintain balance right so there's all of
those things

yeah i think um

following with this quote

from the point of view

of the fep

the life cycle is that which corresponds
to the thing whose integrity is
maintained over evolutionary time

so then the ants nest architecture
is as much recapitulated as is the
casing of the larva

as is the larva itself

so

how will we define life i guess we can
just add that as a good you know just

sort of just something quick to hit at
the end of 32 or of 34.2 just you know
what is life and then um hash rodingers
question been answered

i mean it's been a few years since 2018

and ram said it all

but has it been answered

what is life

1944

um okay

so i think we will uh

conclude there

i hope if you're listening that you

would like to join us in the dot 2

where we'll talk a little more about

this paper and

how we got here and then where we're

going to go

in

22.

anything else dave or blue

okay thanks for the fun convo

talk to you all later